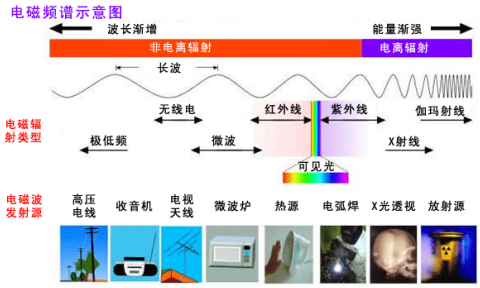
Ch21

$F = \frac{1}{4\pi\epsilon_0}\frac{|q_1 q_2|}{r^2}$ (Coulomb's law: force between two point charges)

$\vec{E} = \frac{\vec{F}_0}{q_0}$ (definition of electric field as electric force per unit charge)

$\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{r}$ (electric field of a point charge)

Mechanical waves 机械波 传播速度 v = $\sqrt{\frac{E}{\rho}}$ (?)



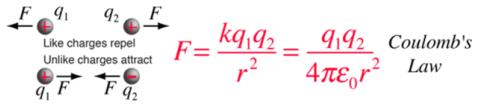
gravitational interactions 引力相互作用

(Universal gravitational force 万有引力 $\vec{F} = -\frac{Gm_1m_2}{r^2}\hat{r}$)

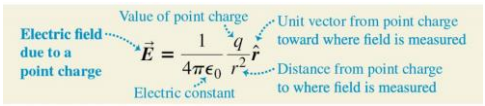
electromagnetic interactions 电磁相互作用

the strong interaction 强相互作用

the weak interaction 弱相互作用



-Electric field due to aa point charge

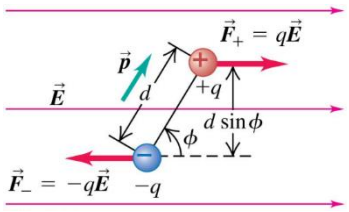


- $\epsilon_0 = 8.854 \times 10^{-12}\ \mathrm{F \cdot m^{-1}}$
- $4\pi\epsilon_0 = k = 9 \times 10^9 \mathrm{N \cdot m^2/C^2}$

spacing(n.)间距

superposition (n.)叠加

-Electric dipole



Electric dipole moment $p = q\,d$

Unit: C·m

Torque for a dipole in an electric field

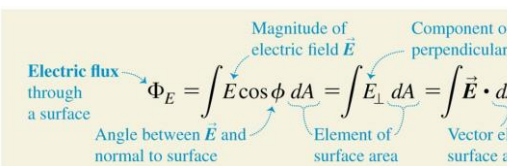
- $\vec{\tau} = \vec{p} \times \vec{E}$

Potential energy for a dipole in an electric field

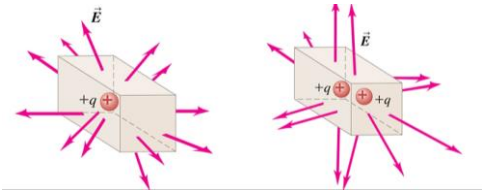
- $U = \vec{p} \cdot \vec{E}$

Ch22

- Electric flux 电通量



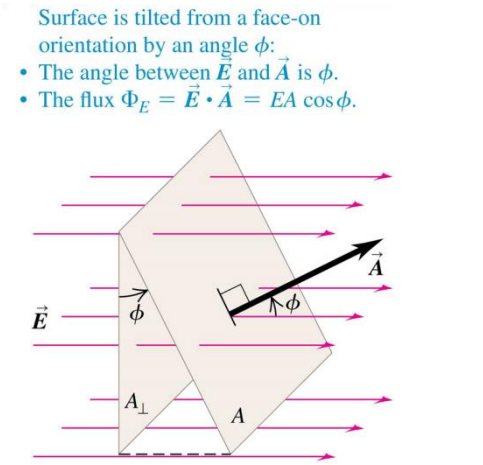
单位: N · m²/C.

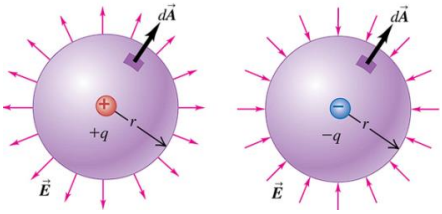


The net electric flux 净电通量

与面内的净电荷 成正比,

is independent of 面的大小 (无关)



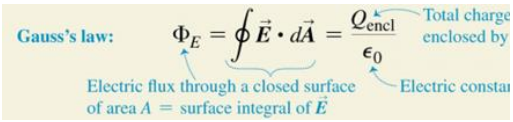


(+) positive (outward) flux 正-向外

(-) negative (inward) flux 负-向内

- Gauss's Law: 高斯定律是表面上 所有点

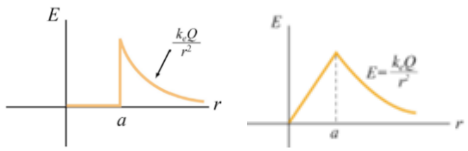
_的电场 与 表面内的_总电荷 _之间的 关系



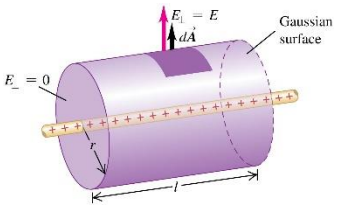
导体内部: E=0 everywhere → Q_{encl}=0 → 电荷在表面

electrostatic conditions: charges not in motion

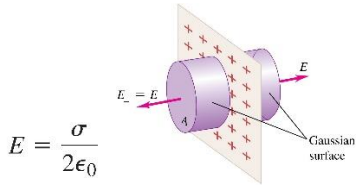
Spherical shell ; Non-Conducting Solid Sphere



Field of a uniform line charge $E = \frac{1}{2\pi\epsilon_0}\frac{\lambda}{r}$

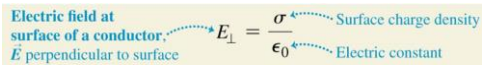


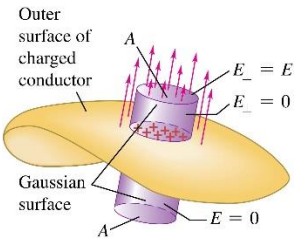
Field of an infinite plane sheet of charge



$E = \frac{\sigma}{2\epsilon_0}$

Field at the surface of a conductor





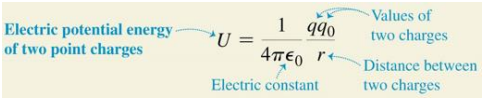
导体上的一片电荷产生的场与仅仅一片电荷产生的场不同: 因为电荷片的两边都有电场, 而导体只有外部有电场

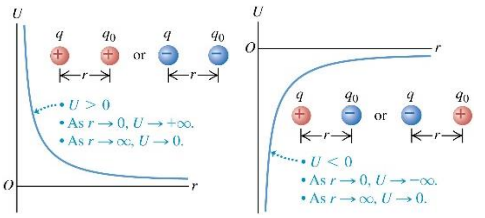
Electrostatic shielding 电磁屏蔽: 导体壳内部 E=0, 电荷正负分别分布于表面左右

内部有空腔的实心导体: 内表面电荷必为内部存在的电荷的相反数

Ch23

- Electric potential energy

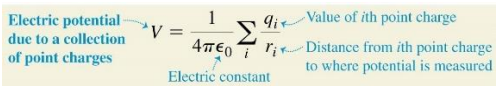




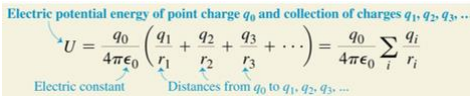
Electric potential $V_{ab} = V_a - V_b$

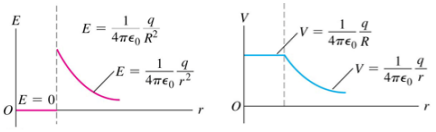
单位电荷从 A 移动到 B 时电力所做的功

The potential due to a collection of point charges:



Electrical potential with several point charges





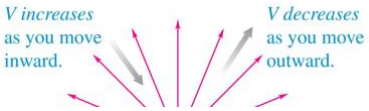
equipotential surface 等势面

导体的表面始终是等电位表面。

导体外的电场总是垂直于导体表面

$U_a - U_b = q\,(V_a - V_b)$

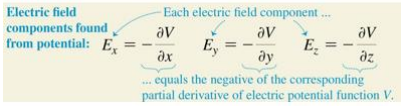
$1\ \mathrm{eV} = 1.602 \times 10^{-19}\ \mathrm{J}$



$$V_a - V_b = -\int_b^a \vec{E} \cdot d\vec{l}$$

1 N/C = 1 V/m

- Potential gradient 电位梯度



Electrostatics
Charge q
Electric force $\vec{\mathbf{F}}_e = k_e \frac{Qq}{r^2}\hat{\mathbf{r}}$
Electric field $\vec{\mathbf{E}} = \vec{\mathbf{F}}_e / q$
Potential energy change $\Delta U = -\int_A^B \vec{\mathbf{F}}_e \cdot d\vec{\mathbf{s}}$
Electric Potential $\Delta V = -\int_A^B \vec{\mathbf{E}} \cdot d\vec{\mathbf{s}}$
Potential function, $V(\infty) = 0$: $V = k_e \frac{Q}{r}$
$\lvert\Delta U\rvert = qEd$, (constant $\vec{\mathbf{E}}$)

Ch24

$C = \frac{Q}{V_{ab}} = \frac{Q}{Ed} = \varepsilon_0 \frac{A}{d}$ (in vacuum)

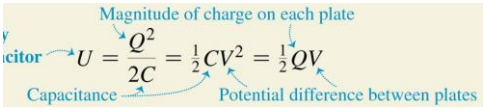
1F = 1C²/N·m = 1C²/J

1V = 1C/J → 1F = 1C/V

1C²/N·m² = 1F/m

→ $\varepsilon_0 = 8.85 \times 10^{-12} F/m$

The potential energy stored in a capacitor:



work to charge a capacitor

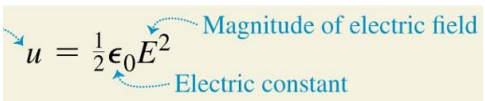
· $dW = v\,dq = \frac{q\,dq}{C}$

$$W_{total} = \int_0^{W_{total}} dW = \frac{1}{C}\int_0^{Q_{final}} q\,dq = \frac{Q_{final}^2}{2C}$$

=the total work done by the electric field on the charge

when the capacitor **discharges 放电**

The energy density:



串联电路 series:

· $V = V_1 + V_2$

· $\frac{1}{C_{eq}} = \frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots$

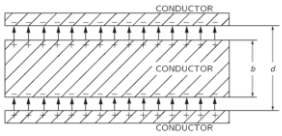
并联电路 parallel:

· $Q = Q_1 + Q_2$

· $\frac{1}{C_{eq}} = \frac{V}{Q} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \cdots$

Electric energy density in a vacuum:

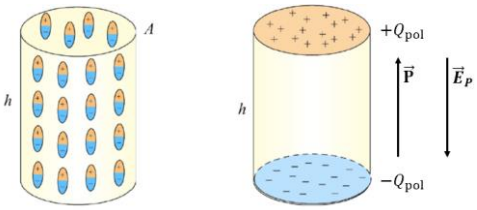
· $u = \text{Energy density} = \frac{\frac{1}{2}CV^2}{Ad} = \frac{1}{2}\varepsilon_0 E^2$



$$V = \frac{\sigma}{\epsilon_0}(d-b) \qquad C = \frac{\epsilon_0 A}{d[1-(b/d)]}$$

$$\vec{\mathbf{P}} = \frac{1}{\text{Volume}}\sum_{i=1}^{N}\vec{\mathbf{p}}_i \qquad P = \frac{Np}{Ah}$$

$$Q_{\text{pol}} = \frac{Np}{h} \qquad \sigma_{\text{pol}} = \frac{Q_{\text{pol}}}{A} = \frac{Np}{Ah} = P$$



The dielectric constant K

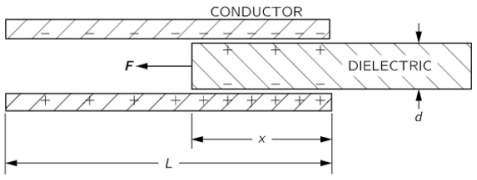
· $\varepsilon = K\,\varepsilon_0$ (in vacuum K=1)

Dielectrics 电介质: nonconducting material

Dielectric breakdown 介电击穿

dielectric strength: the maximum electric field……

- The force on a dielectric thin film



$$Q = \frac{\kappa\epsilon_0 V}{d}xW + \frac{\epsilon_0 V}{d}(L-x)W, \qquad C = \frac{\epsilon_0 W}{d}(\kappa x + L - x).$$

$$F_x = -\frac{\partial U}{\partial x} = +\frac{V^2}{2}\frac{\partial C}{\partial x}. \qquad F_x = \frac{V^2}{2}\frac{\epsilon_0 W}{d}(\kappa - 1).$$

Ch25

电流 · $I = \frac{dQ}{dt} = n\,|q|\,v_d\,A$

• $v_d = \tau\,a = qE\tau/m$

n: Concentration of moving charged particles 移动带电粒子的浓度

|q|: Charge per particle 每个粒子的电荷

- The total current I : is found by adding up the currents due to each kind of charged particle.

-Current density 电流密度

· $\vec{J} = n\,q\,\vec{v_d}$

矢量 电流密度 J 的方向 始终 与 矢量 电场 E(?) 的方向 相同，无论电荷载流子的符号是什么

-Resistivity (ρ) 电阻率

·$\rho = \frac{E}{J}$ (电场大小/电流密度大小)

Conductivity (σ) 电导率

·$\sigma = \frac{1}{\rho}$

The **resistance 电阻** of a conductor:

· $R = \frac{\rho L}{A}$

· $V = IR$

电流从 较高电位 流向 较低电位 (电流流向与电势)

在给定温度下，电流与电压成正比

-Nonohmic resistors 非欧姆电阻器

电压与电流的关系可能不是 成正比的，

对于 电流的两个方向(!!)，电压与电流的关系可能不同。

semiconductor diode 半导体二极管

diode (n.)二极管

温度 T↑，电阻率 ρ↑

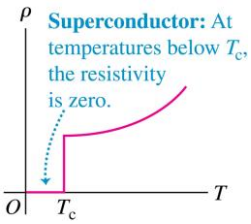
· $\rho(T) = \rho_0[1 + \alpha(T - T_0)]$

α: Temperature coefficient of resistivity

-Thermistor (n.)热敏电阻

温度 T↑，石墨的电阻率 ρ↓

-Superconductivity (n.)超导性



phase transition 相变

- Electromotive force (emf) (ε) 电动势

1V = 1J/C

Terminal voltage 端电压

·$V_{ab} = \varepsilon - Ir$

Power 功率·$P = V_{ab}I$

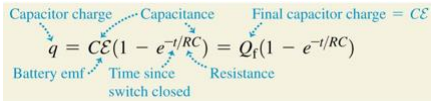
Ch26

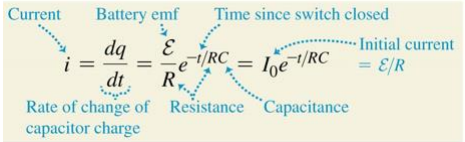
-Kirchhoff's rules 基尔霍夫定律

·$\sum I_{in} = \sum I_{out}$

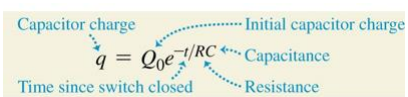
·$\sum_{closed\ loop} \Delta V = 0$

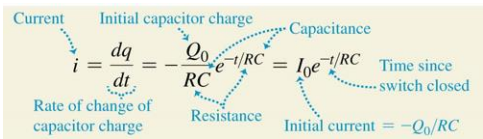
电容器充电时：





电容器放电时：





$$P_{\text{avg}} = \langle I(t)^2 R\rangle \quad \text{where } \langle\cdots\rangle \text{ denotes the mean of a function}$$
$$= \langle I(t)^2\rangle R \quad \text{(as } R \text{ does not vary over time, it can be factored out)}$$
$$= I_{\text{RMS}}^2 R \quad \text{by definition of root-mean-square}$$